

FOUNDATION PROGRAM IN LITERACY, NUMERACY AND SKILLS

NUMERACY

TEACHER'S GUIDE: GRADE 8

Lesson 19 & 20: Pythagoras theorem

A. To discover and explore the Pythagoras theorem and its application through activities.

B. Competency, Element and Performance Criteria:

C.2.2: Application of geometry in practical situations and trades

Element 5: Use Pythagoras Theorem

Performance Criteria:

Range – Level 1: Use Pythagoras theorem to find an unknown side of a right-angled triangle.

Range – Level 2: Apply Pythagoras theorem in solving word problems.

C. Purpose of activity:

This activity will help the students understand the concept and explore how areas of squares relate to each other and show students how the theorem is useful in fields like construction, navigation and design.

D. Learning Outcomes:

By the end of the lesson, students should be able to:

- Determine the Pythagorean relationship and prove the Pythagoras theorem using manipulatives.
- Use the Pythagoras theorem to find the length of the hypotenuse of a right angled triangle
- Make students grasp that in a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides
- Use the Pythagoras theorem to find the length of a side of a right- angle triangle.

E: Resources and materials:

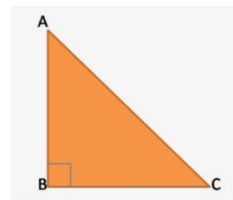
- Graph paper
- Scissors
- Worksheets

F: Teaching trajectories/ Implementation guide:

Activity 1: Introduction (Prior knowledge)

The teacher asks the students “what they know about right angle triangles”

Explain the parts: Right angles, two shorter sides and longest side.



Record all the responses on the whiteboard.

The teacher can ask the following questions:

- What type of angle does a right angle triangle always have?
- How do we call the longest side in a right angle triangle?
- What is the hypotenuse?

The students are then asked to identify the hypotenuse in student’s worksheet (Question 1)

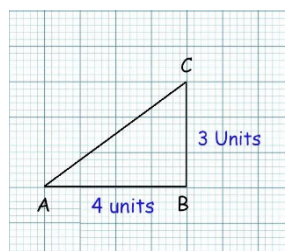
Activity 2: Introduction of Pythagoras theorem

Pythagoras theorem is used only for right angle triangles



- Students will be divided into groups and provided with materials to explore the relationship between the squares of the lengths in a right angle triangle and derive the formula for the Pythagoras Theorem. After providing instructions, the teacher will listen and observe students as they work on task and ask questions if necessary.

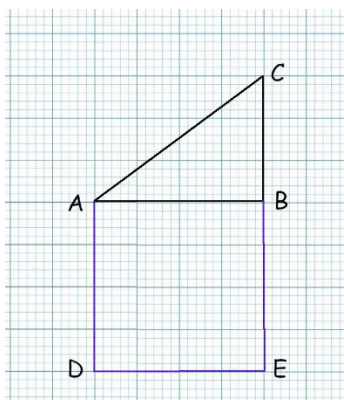
Step 1 – Use graph paper triangle whose base (AB) is 4 perpendicular (BC) is 3 units.



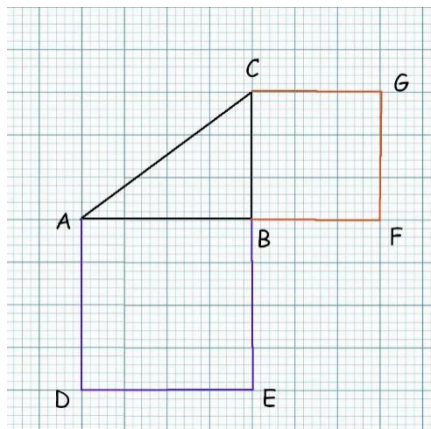
and draw a right angle units and one side

Step 2 – On the graph paper, draw squares corresponding to the 3 sides of the triangle ABC.

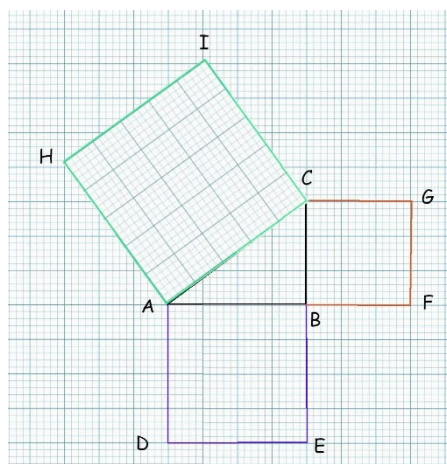
Draw a square ABED at base (AB) of side 4 units.



Draw a square BCGF of side 3 units.

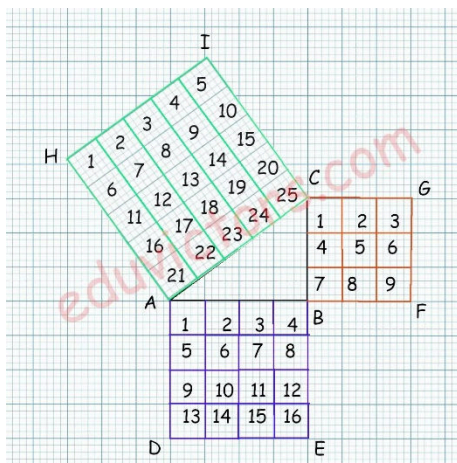


Take a separate square ACHI of



graph paper and draw a side equal to AC.

Step 3 – Now, count the number of unit squares in the 3 squares obtained (ABED, BCGF and ACHI). What do you observe? Try to establish a relationship among the areas of the 3 squares and then one among the 3 sides.



Area of ABED = 16 square units

Area of BCGF = 9 square units

Area of ACHI = 25 square units

What is the relationship between the areas of the squares?

➤ $16 + 9 = 25$

What numbers to the square are equal to 25, 16 and 9?

➤ $4^2 + 3^2 = 5^2$

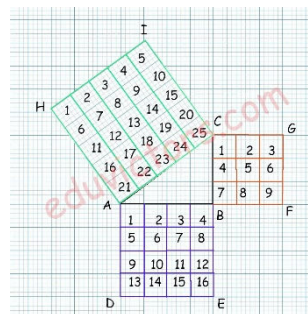
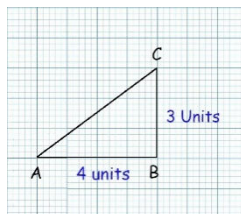
Step 4 – Compare the results with the right angle triangle given at first and derive the formula for the Pythagoras Theorem.

➤ $4^2 + 3^2 = 5^2$

What do these numbers represent?

➤ $AB^2 + BC^2 = AC^2$ or

➤ $AC^2 = AB^2 + BC^2$

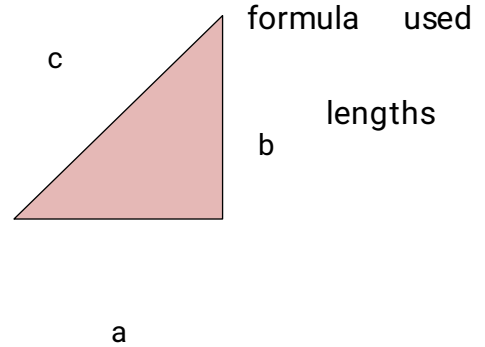


We conclude that the square of length of the hypotenuse (AC^2) is equal to the

sum of the squares of the length of the other two sides (AB^2 , BC^2).

That is what the Pythagoras Theorem states. The formula used for the Pythagoras theorem is:

(hypotenuse)² = sum of the squares of the lengths of the other two sides i.e. $c^2 = a^2 + b^2$



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